SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Smart Visitor Pass Management System

Requirements Baseline v1.0, updated to v1.1 via Change Request CR-001 (Manual Entry / Exit Fallback)

DOCUMENT INFORMATION

Project	Smart Visitor Pass Management System
Document	Software Requirements Specification (SRS)
Version	1.1 (baseline v1.0 + CR-001)
Baseline	Requirements Baseline v1.0, revised under CR-001 to v1.1
Development Approach	Incremental Development with Controlled Change Management
Technology	HTML, CSS, JavaScript, PHP, MySQL
Version Control	Git and GitHub
Status	Proposed for Review
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Date	29/08/2026

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of the Smart Visitor Pass Management System, a web-based application for digitally registering, tracking, and monitoring visitors at an organization's premises.

The document establishes the agreed requirements that will be used as the reference for system analysis, system design, database design, software development, testing, release activities, and Software Change Management.

The requirements contained in this document represent the initial proposed scope of the system. Once reviewed and approved, this document became the Requirements Baseline v1.0. A subsequent Change Request, CR-001, was raised to introduce a manual entry/exit fallback (using the visitor's ID number when the QR pass cannot be scanned), and the resulting revised requirements are recorded in Section 23 of this document as Requirements Baseline v1.1.

1.2 Project Background

Most institutions still manage visitors using a manual, paper-based register at the security desk. This approach is slow, prone to human error, and makes it very difficult to search past visitor records. There is no real-time way to know how many visitors are currently inside the premises, and exit times are often not recorded at all, leaving visit records incomplete. There is also no unique verifiable pass for each visitor, making identity confirmation fully dependent on the security guard's manual judgement.

The Smart Visitor Pass Management System is intended to replace the manual register with a digital, QR-code based platform where security staff can register a visitor's details, generate a unique QR-coded pass, and scan the pass to record entry and exit, while administrators get a live view of everyone currently on the premises.

2. Problem Statement

- Visitor details are recorded manually in a paper register, which is slow and error-prone.
- There is no real-time way to know how many visitors are currently on the premises.
- Exit times are frequently not recorded, leaving visit records incomplete.
- There is no unique, verifiable pass for each visitor, so identity confirmation depends entirely on the security guard's judgement.
- Searching past visitor records is difficult and time-consuming with a paper-based system.

3. Project Objectives

- To develop a web-based system for registering visitor details digitally.
- To automatically generate a unique QR-coded pass for every visitor.
- To enable QR-code scanning at the gate for recording entry and exit times.
- To provide a live dashboard showing visitors currently inside the premises.
- To maintain a searchable history of all visitor records by name, host, or date.
- To reduce manual record-keeping errors and improve data accuracy.
- To provide an administrator with oversight of staff, visitors, and visit records through an admin dashboard.
- To allow entry and exit to still be recorded when the QR pass cannot be scanned, through a manual entry feature using the visitor's ID number (introduced under CR-001).

4. Project Scope

4.1 In Scope for Version 1.0

Authentication

- Secure login for Admin and Security Staff (role-based access)
- Logout
- Credential validation
- Secure password storage

Visitor Management

- Register visitor details (name, phone, email, purpose, host details)
- Edit / view registered visitor details
- Automatic QR-coded visitor pass generation

Entry / Exit Tracking

- QR code scanning at the gate to record entry time
- QR code scanning at the gate to record exit time

Dashboard & History

- Live dashboard of visitor statistics and current occupancy
- Searchable visitor history with name, host, and date-range filters

Administration

- Admin dashboard to view all registered staff accounts
- Admin dashboard to view all visitors and their visit records

Technical Scope

- HTML, CSS, JavaScript frontend
- PHP backend
- MySQL database
- Git and GitHub version control

4.2 Out of Scope for Version 1.0

- Manual entry/exit without a QR scan (raised separately and approved under CR-001 — see Section 23)
- Visitor self-registration before arrival (visitors are registered by security staff only)
- Mobile application (native)
- SMS / email pass delivery to visitors
- Advanced role-based access control beyond Admin and Security Staff
- Production-scale deployment

5. Stakeholders

Stakeholder	Role	Interest / Responsibility
Security Staff	System User	Register visitors, generate passes, scan visitors in and out
Administrator	System User	Manage staff accounts, monitor dashboard, review visitor history
Visitor	End User (non-login)	Provides details for registration; carries and presents the QR pass
Organization Management	Governance	Monitor security posture and visitor activity on premises
Project Team	Development Team	Analyze, design, develop, test, and maintain the system
Lecturer / Supervisor	Academic Supervisor	Monitor project progress and approve Change Requests

6. User Roles

6.1 Administrator

The administrator manages the overall system through an admin dashboard. The administrator shall be able to:

- Log in and log out
- View the list of all registered security staff accounts
- View all registered visitors and their pass status
- View all entry and exit records, including how each was captured
- Search and filter visitor history by name, host, or date

6.2 Security Staff

Security staff register visitors and scan passes at the gate. Security staff shall be able to:

- Log in and log out
- Register a new visitor and generate a QR-coded pass
- Scan a visitor's QR pass to record entry
- Scan a visitor's QR pass to record exit
- View the live dashboard of visitors currently on the premises

6.3 Visitor

A visitor does not log in to the system. The visitor provides their details to security staff for registration and receives a QR-coded pass, which is presented at the gate for scanning on entry and exit.

7. Functional Requirements (Version 1.0)

Functional requirements describe what the system shall do.

7.1 Authentication Requirements

FR-01 – Staff Login
The system shall allow a registered Administrator or Security Staff member to log in using valid credentials.

FR-02 – Login Validation
The system shall validate the credentials provided and display an appropriate error message if invalid.

FR-03 – Staff Logout
The system shall allow an authenticated staff member to log out of the system.

7.2 Visitor Registration & Pass Requirements

FR-04 – Register Visitor
The system shall allow authenticated security staff to register a new visitor with name, phone number, email, purpose of visit, and host details.

FR-05 – Generate QR Pass
The system shall automatically generate a unique QR-coded pass for each registered visitor.

FR-06 – View / Edit Visitor Details
The system shall allow authenticated security staff to view and edit a registered visitor's details before entry is recorded.

7.3 Entry / Exit Tracking Requirements

FR-07 – Scan Entry
The system shall allow security staff to scan a visitor's QR pass at the gate to record the visitor's entry time.

FR-08 – Scan Exit
The system shall allow security staff to scan a visitor's QR pass at the gate to record the visitor's exit time.

FR-09 – Prevent Duplicate Entry
The system shall not allow a visitor to be scanned for entry again while they already have an open (unexited) visit.

7.4 Dashboard & History Requirements

FR-10 – Live Occupancy Dashboard
The system shall provide a live dashboard showing the number and details of visitors currently on the premises.

FR-11 – Search Visitor History
The system shall allow a user to search and filter past visitor records by visitor name, host, or date range.

7.5 Administration Requirements

FR-12 – Admin Dashboard
The system shall provide an administrator with a dashboard showing all registered staff, all visitors, and all entry/exit records.

FR-13 – Manage Staff Accounts
The system shall allow an administrator to add, view, or deactivate security staff accounts.

8. Non-Functional Requirements

8.1 Security

NFR-01 – Password Security
The system shall store passwords using secure password hashing rather than plain text.

NFR-02 – Authentication
Protected system functions shall require the staff member to be authenticated.

8.2 Usability

NFR-03 – User Interface
The system shall provide a simple, understandable, and consistent user interface for security staff working at the gate.

NFR-04 – Validation Messages
The system shall provide clear messages when input is invalid or an operation cannot be completed.

8.3 Reliability & Performance

NFR-05 – Response Time
Commonly used pages, including the scan screen, should respond within approximately three seconds under normal conditions.

NFR-06 – Data Integrity
The system shall maintain the integrity and consistency of stored visitor and visit record data.

8.4 Portability & Responsiveness

NFR-07 – Browser Compatibility
The web application should operate correctly on commonly used modern web browsers.

NFR-08 – Responsive Interface
The user interface should adapt to common desktop, tablet, and mobile screen sizes, including devices used at the gate for scanning.

9. User Stories

ID	Title	Story
US-01	Staff Login	As a staff member, I want to log in, so that I can use the functions available to me.
US-02	Register Visitor	As a security staff member, I want to register a visitor's details, so that a pass can be generated for them.
US-03	Generate Pass	As a security staff member, I want the system to generate a QR pass automatically, so that the visitor has a unique verifiable pass.
US-04	Scan Entry	As a security staff member, I want to scan a visitor's pass at entry, so that the entry time is recorded accurately.
US-05	Scan Exit	As a security staff member, I want to scan a visitor's pass at exit, so that the exit time is recorded accurately.
US-06	Live Dashboard	As an administrator, I want to see a live dashboard of who is currently on the premises, so that I have real-time visibility.
US-07	Search History	As an administrator, I want to search past visitor records by name, host, or date, so that I can quickly find a specific visit.
US-08	Admin Dashboard	As an administrator, I want to view all staff, visitors, and visit records, so that I can monitor and manage the system.

10. Acceptance Criteria

AC-01 – Staff Login Given a registered staff member is on the login page, when valid credentials are entered, then the staff member shall be authenticated and redirected to the dashboard. Invalid credentials shall show an error.

AC-02 – Register Visitor Given authenticated security staff, when valid visitor details are submitted, then the visitor shall be saved and a unique QR pass generated.

AC-03 – Scan Entry Given a visitor has a valid QR pass and no open visit, when the pass is scanned at the gate, then the entry time shall be recorded and the visitor shall appear on the live dashboard.

AC-04 – Scan Exit Given a visitor has an open (unexited) visit, when their pass is scanned at the gate, then the exit time shall be recorded and the visitor shall be removed from the live dashboard.

AC-05 – Prevent Duplicate Entry Given a visitor already has an open visit, when their pass is scanned again for entry, then the system shall reject the scan and display an appropriate message.

AC-06 – Admin Dashboard Given an administrator is logged in, when the dashboard is opened, then all staff, visitors, and visit records shall be displayed.

11. MoSCoW Prioritization

Priority	Requirements
Must Have	Staff Login/Logout, Register Visitor, Generate QR Pass, Scan Entry, Scan Exit, Live Dashboard, Admin Dashboard
Should Have	Visitor History Search, Improved Validation, User Feedback Messages
Could Have	Visitor Photo Capture, Date-Range Filters
Won't Have in v1.0	Manual Entry / Exit Fallback (introduced later via CR-001)

12. Product Backlog

ID	Item	Priority	Related Req.	Status
US-01	Staff Login / Logout	Must	FR-01, FR-02, FR-03	Planned
US-02	Register Visitor	Must	FR-04	Planned
US-03	Generate QR Pass	Must	FR-05	Planned
US-04	Scan Entry	Must	FR-07	Planned
US-05	Scan Exit	Must	FR-08	Planned
US-06	Live Dashboard	Must	FR-10	Planned
US-07	Search History	Should	FR-11	Planned
US-08	Admin Dashboard	Must	FR-12, FR-13	Planned
—	Manual Entry / Exit Fallback	Won't Have (v1.0) → Approved (v1.1)	See CR-001	Future / In Progress

13. Use Cases

Use Case ID	Use Case	Primary Actor
UC-01	Login	Staff (Admin / Security)
UC-02	Register Visitor	Security Staff
UC-03	Generate QR Pass	System (triggered by Security Staff)
UC-04	Scan Entry	Security Staff
UC-05	Scan Exit	Security Staff
UC-06	View Live Dashboard	Administrator / Security Staff
UC-07	Search Visitor History	Administrator

14. Business Rules

- BR-01 – Only authenticated security staff or administrators may register visitors or scan passes.
- BR-02 – A QR-coded pass shall be unique to a single visit and shall not be reused for a different visit.
- BR-03 – A visitor cannot be scanned for exit unless they have a valid, prior recorded entry with no exit yet.
- BR-04 – Passwords must not be stored in plain text.
- BR-05 – Each staff account shall have a unique username or email address.

15. Data Requirements

15.1 Staff Information

- Staff ID
- Name
- Email / Username
- Password (hashed)
- Role (Admin / Security Staff)

15.2 Visitor Information

- Visitor ID
- Name
- Phone Number
- Email
- Purpose of Visit
- Host Name / Department
- QR Code (pass reference)
- Pass Status (Active / Used / Expired)

15.3 Visit Record Information

- Visit ID
- Visitor ID
- Entry Time
- Exit Time
- Recorded By (Staff ID)

16. System Constraints

- 1. The application will be developed as a web-based system.
- 2. HTML, CSS, and JavaScript will be used for the frontend.
- 3. PHP will be used for the backend.
- 4. MySQL will be used as the database.
- 5. Git and GitHub will be used for version control.
- 6. A device with a camera (e.g., laptop, tablet, or phone) is required at the gate for QR scanning.
- 7. Version 1.0 focuses on core registration, pass generation, and QR-based entry/exit tracking.
- 8. New requirements after baseline approval must follow the Change Request process.

17. Assumptions

- Security staff will have access to a device with a camera for scanning QR passes.
- Visitors will be physically present at the gate for their pass to be scanned.
- The development environment will use XAMPP (Apache, PHP, MySQL).
- Requirements may evolve after Version 1.0 and will be managed through Software Change Management.

18. Requirements Traceability Matrix

Requirement	User Story	Acceptance Criteria
FR-01/02/03	US-01	AC-01
FR-04/05	US-02, US-03	AC-02
FR-07	US-04	AC-03
FR-08/09	US-05	AC-04, AC-05
FR-10	US-06	AC-03, AC-04
FR-11	US-07	—
FR-12/13	US-08	AC-06

19. Requirements Review

Q: Is manual entry/exit without scanning the QR pass included in Version 1.0?

Decision: No. It was not part of the original proposal and is introduced later through CR-001 (see Section 23).

Q: Can visitors register themselves online before arrival?

Decision: No. In Version 1.0, all visitors are registered by security staff at the point of arrival.

Q: Can a security staff account access the admin dashboard?

Decision: No. The admin dashboard is restricted to the administrator role.

20. Version 1.0 Requirements Baseline

After stakeholder and supervisor review, the requirements contained in Sections 1–19 of this document were approved and established as Requirements Baseline v1.0 for the Smart Visitor Pass Management System.

After baselining, any new requirement is submitted through the Software Change Management process using a formal Change Request (CR), rather than being added directly to the approved scope.

21. Configuration and Version Control

Item	Value
Baseline Name	Requirements Baseline v1.0
Git Tag	requirements-v1.0
Status	Approved after Sign-off
Baseline Date	_____

22. Change Management Process

Once Version 1.0 is baselined, new requirements are handled as follows:

Customer / Supervisor Request → Change Request (CR) → Impact Analysis → Change Approval → Requirements Update → Design Update → Database Update → Backend Update → Frontend Update → Testing → Documentation Update → New Version / Release

23. Change Request CR-001 – Manual Entry / Exit Fallback

Following approval of Requirements Baseline v1.0, it was identified that the system should not depend entirely on QR scanning: if a visitor misplaces or damages their QR pass, or if the scan otherwise fails, security staff still need a way to record entry and exit. This section documents CR-001 in full and records the resulting Requirements Baseline v1.1.

23.1 Change Request Form

Field	Details
CR ID	CR-001
Title	Introduce Manual Entry / Exit via Visitor ID Number
Raised By	Supervisor / Project Team, during project review
Description	Add the ability for security staff to manually record a visitor's entry or exit by entering the visitor's ID number, when the QR pass cannot be scanned (lost, damaged, or unreadable).
Reason	The system should not fully depend on a successful QR scan; visitors may misplace or damage their pass, and the visit must still be logged accurately.
Affected Baseline	Requirements Baseline v1.0
Priority	Should Have
Status	Approved

23.2 Impact Analysis

Area	Impact
Requirements	New functional requirements FR-14 to FR-17 added (Section 23.3).
Database Design	The visit_records table requires an entry_method / exit_method field (QR / Manual) and a recorded_by (staff) reference.
Backend	New logic to look up a visitor by ID number, validate that a valid pass exists, and record a manual entry or exit.
Frontend	New "Manual Entry" option on the scan screen with an input field for the visitor's ID number.
Testing	New test cases required to confirm manual entry/exit works correctly and that invalid ID numbers are rejected.
Schedule / Effort	Low to moderate — feature is additive and does not change the existing QR-scan workflow.

23.3 Updated Functional Requirements (Version 1.1)

FR-14 – Manual Entry via Visitor ID
The system shall allow authorized security staff to record a visitor's entry manually by entering the visitor's ID number, when the QR pass cannot be scanned.

FR-15 – Manual Exit via Visitor ID
The system shall allow authorized security staff to record a visitor's exit manually by entering the visitor's ID number, when the QR pass cannot be scanned.

FR-16 – Validate Visitor ID for Manual Entry
The system shall validate that the entered visitor ID number corresponds to a registered visitor with a valid pass before recording a manual entry or exit, and shall display an error if the ID is invalid or has no matching visitor.

FR-17 – Log Entry / Exit Method
The system shall record whether each entry or exit was captured via QR scan or manual entry, along with the staff member who performed it, for audit purposes.

23.4 Updated User Stories & Acceptance Criteria

ID	Story
US-09	As a security staff member, I want to manually record a visitor's entry using their ID number, so that I can still log the visit if the QR pass cannot be scanned.
US-10	As a security staff member, I want to manually record a visitor's exit using their ID number, so that the visit record remains complete even without a successful scan.
US-11	As an administrator, I want to see whether an entry or exit was recorded by scan or manual entry, so that I can audit unusual records.

AC-07 – Manual Entry Given a visitor's QR pass cannot be scanned, when a security staff member enters a valid visitor ID number and confirms entry, then the visitor's entry time shall be recorded and marked as "Manual".

AC-08 – Manual Exit Given a visitor's QR pass cannot be scanned and the visitor has an open visit, when a security staff member enters the visitor's ID number and confirms exit, then the exit time shall be recorded and marked as "Manual".

AC-09 – Invalid ID Handling Given a security staff member enters a visitor ID number that does not exist or has no active entry, when they attempt to record a manual entry or exit, then the system shall show an error message and shall not record the action.

23.5 Updated Business Rules

- BR-06 – A manual entry or exit shall only be permitted by an authenticated security staff or administrator account.
- BR-07 – Every manual entry or exit action shall be logged with the staff member's identity and timestamp for audit purposes.

23.6 Updated Data Requirements — Visit Record Information

- Entry Method (QR Scan / Manual)
- Exit Method (QR Scan / Manual)
- Recorded By (Staff ID) — already present, now also applies to manual actions

23.7 Updated MoSCoW Prioritization

Priority	Requirements
Must Have (v1.1)	All Version 1.0 Must Have items
Should Have (v1.1)	Manual Entry / Exit Fallback (FR-14 to FR-17) — approved under CR-001
Could Have	Visitor Photo Capture, Date-Range Filters

24. Version 1.1 Requirements Baseline (Post-CR-001)

Following impact analysis and approval of CR-001, the requirements in Section 23.3 (FR-14 to FR-17) are incorporated into the system scope. The combined Version 1.0 requirements together with the CR-001 additions constitute Requirements Baseline v1.1 for the Smart Visitor Pass Management System.

Item	Value
Baseline Name	Requirements Baseline v1.1
Change Request	CR-001 – Manual Entry / Exit Fallback
Git Tag	requirements-v1.1
Status	Approved after Sign-off
Baseline Date	_____

25. Version History

Version	Date	Description	Status
0.1	_____	Initial requirements draft	Draft
1.0	_____	Requirements baseline established	Approved
1.1	_____	CR-001 (Manual Entry / Exit Fallback) analyzed and merged into baseline	Approved

26. Approval

By approving this document, the relevant stakeholders confirm that the requirements defined, including the Version 1.1 update introduced under CR-001, represent the agreed scope of the Smart Visitor Pass Management System.

Role	Name	Signature	Date
Student / Project Team	_____	_____	_____
Lecturer / Supervisor	_____	_____	_____